

The Fuzzy Subjective Structure Model (FSSM)

Evaluating Causal Claims for Non-Deterministic Inference on Subjective Constructs

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The Generalization Impasse in LIS

- LIS possesses rich theoretical frameworks, but empirical studies are bound and limited by methods oriented toward broad **population-wide generalization**.
- A typical information behavior (IB) in context research project requires its own **specific design and methodological development**.
- The full potential of in-context research is not in chasing sweeping averages, but in **supporting decision-making based on local evidence**.
- We need dedicated designs that evaluate local evidence and context-specific configurations rather than default generalization.

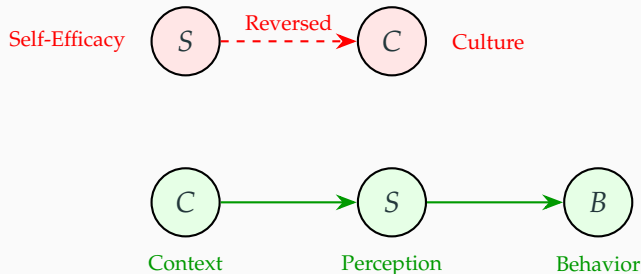
Subjective Structures & Effects of Causes

- **Subjectivity is the very area where the effect of cases is located.**
- The FSSM captures how the effects of a cause—whether **adjacent** (proximal interaction) or **coercive** (structural policies/norms)—reside within the subjective structure of human perception rather than acting as deterministic drivers.
- Environmental and institutional factors do not directly force behavior; instead, their impact is filtered through how individuals subjectively perceive and experience affordances and constraints.

Beyond the “Average” Human

- Traditional regression models rely on mean estimates that represent a vague abstraction of a “**perfect human being**”—which none of us is.
- Such approaches discard the individual **autonomy of thoughts**.
- FSSM shifts focus to **subjectivity mapping** to assess:
 - **Typicality**: Regularities that hold for nearly all cases in a group.
 - **Necessity**: Non-compensatory system bottlenecks (critical gatekeepers).
 - **Probability**: Individual likelihood of behaving as an effect of context causes.
- We examine how far or close individuals are toward modeled desirable thresholds, rather than calculating aggregate averages.

Arrow-Reversal & Context-First Skeleton



- Theory dictates: **Context** → **Perception** → **Behavior**.
- Conventional regression often violates this for statistical convenience (e.g., Self-Efficacy driving Culture).
- FSSM enforces the theoretical skeleton: Culture/Management act as upstream enabling context conditions.

The Four Causal Claims of FSSM

1. **Typical Sufficiency (α):** *“If X is high, Y is typically high.”* (Evaluates if general policies usually suffice, permitting rare exceptions).
2. **Typical Necessity (ϵ):** *“If Y is high, X is rarely low.”* (Identifies system bottlenecks or gatekeepers).
3. **Probabilistic Sufficiency (Δ):** Probability uplift on a balanced pseudo-population after covariate adjustment.
4. **Probabilistic Necessity (β):** *“If X is low, high Y is unlikely.”* (Converts bottlenecks into counterfactual risk statements).

Dual Logic: Latent vs. Composite Route

- **Route A (Latent Variable Route - PLS-SEM + NCA):** Extends latent structural models. Achieves both population generalization (structural paths) and subjective structure explorations (NCA bottlenecks).
- **Route B (Raw Composite Route):** Bypasses latent modeling demands using raw composites. Enables a case-level “**finding people**” approach.
- Allows us to study teams, even small teams, more precisely regarding their local needs and behaviors in context based on local evidence.

Fuzzy Concept & Subjectivity Mapping

- Maps continuous latent and composite scores onto a fuzzy membership interval $[0, 1]$ relative to empirical thresholds and bottlenecks.
- $\mu_S(x)$: Degree of membership in the necessary context bottleneck.
- $\mu_Y(y)$: Degree of outcome achievement.
- Honors individual autonomy of thoughts by displaying how close or far individuals are from thresholds, rather than using binary cutoffs.

Thank You for Your Attention

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FSSM Project & Updates on OSF:

<https://osf.io/b8maq/>